

Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

L-3/T-1

January 2025

CSE 311: Data Communication

Time: 25 minutes

Marks: 20

Student Name: _____

Student ID: _____

1. Determine the resulting transmitted waveform or spectrum in frequency domain for the following modulation schemes. Do **not** show any calculations. (10)
- (a) DSB-SC

(2)
- (b) DSB + C

(2)
- (c) SSB

(2)
- (d) VSB

(2)
- (e) SSB + C

(2)

2. Why would a TV system choose VSB rather than pure SSB? (2)
3. Explain how phase mismatch or poor carrier synchronization in QAM can cause symbol errors. (2)
4. Why does envelope detection fail when the instantaneous amplitude becomes negative? (2)
5. What difficulty arises in generating SSB using selective filtering when $f_c \gg B$? How can we overcome this problem? (2)
6. Why DSB receiver might be cheaper to build than DSB-SC receiver? (2)